

My Specific Contributions

I worked with Zhuo Chen and Jie Yu Huang participated in compiling and formatting the LaTeX document, ensuring the layout of the final report. I made the initial research and screening of relevant academic literature.

By reading the vehicle owner's manual, I worked with Zhuo Chen and Jie Yu to identified key processes in the BCW/BCA system. We completed the data flow diagram and the corresponding system description, continuously updating it until the final version.

I updated the first draft of the "STRIDE Analysis for Two Systems" section. I systematically applied the STRIDE model to both "Human-Computer Interaction" and "System-Hardware Interaction," generating a comprehensive list of six threats for each boundary.

I conducted an in-depth analysis of the "Privilege Escalation" threat. This included: Writing a detailed STRIDE threat description, conceiving the complete attack chain from physical access to the OBD-II interface to ultimate control of the HMI, and drawing the corresponding attack tree.

Contributions from others

- Jie Yu Huang: is responsible for one of attack trees and threats
- Sanika Pravin Mahamulkar: is responsible for the introduction of report
- Tailai Liu: is responsible for the mitigation and stride overall
- Tianju Yang: is responsible for the first vision of DFD
- Zhuo Chen: is responsible for one of attack trees and threats
- YI LI, Zhuo Chen and Jie Yu Huang: are responsible for the update of the entire project

Group Dynamic

At the start of the project, we met on Saturday to select the blind spot detection system and divide up the work. By Monday, we had completed the first drafts of each component. After receiving feedback from our advisors on Tuesday, Zhuo Chen, Jie Yu Huang, and I made significant revisions to the DFD, threat analysis, and attack tree. We also refined and improved various aspects of the project. In the final stages, Jie Yu Huang, Tailai Liu, Tianju Yang, and Zhuo Chen jointly approved the final version and submitted it.

Knowledge and Skills, I Learned

This project marked my first systematic application of STRIDE theory to a complex, real-world system. By drawing a DFD, I mastered how to transform an abstract model into a structured security analysis tool. Drawing an attack tree was my greatest skill acquisition. It allowed me to think from an attacker's perspective, breaking down the vague goal of privilege escalation into a series of specific, actionable technical steps. The ultimate impact of a successful privilege escalation attack might not be data theft, but rather the failure of vehicle safety features, directly threatening personal safety.